Jackie (Jiaqi) Zou

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Projects

High-Performance Task Scheduler / Simulation Engine Jan 2025 – Apr 2025
(C++, Python, Algorithms, Data Structures) GitHub
- Designed a task scheduling engine using priority queues, heaps, and graph algorithms for optimized resource allocation under constraints
- Improved throughput by 30% via cache-efficient data structures and reduced memory allocations
- Simulated concurrent execution scenarios to validate correctness under high-load contention

Next-Gen Inventory & Operations Backend May 2025 – Aug 2025
(Java, SQL, RESTful API, FastAPI, Celery) GitHub
- Applied prior concurrency and systems principles to build backend supporting inventory, orders, and users with ACID-compliant transactions
- Designed RESTful APIs with clear service boundaries to decouple business logic and external integrations
- Reduced manual operations by 50% via asynchronous workflow automation (Celery workers)
- Structured service-layer architecture to isolate transaction logic and ensure safe concurrent updates

Smart Analytics & Task Management Platform Sep 2025 – Feb 2026
(Python, FastAPI, SQL, Pandas, PyTorch, Streamlit, React) GitHub
- Extended prior backend systems into a data-driven platform integrating APIs, ETL pipelines, and analytics
- Built ETL pipelines for structured/unstructured data, enabling automated reporting and model consumption
- Optimized database queries and APIs for real-time updates under concurrent usage
- Developed dashboards (React + TailwindCSS) to visualize system outputs and simulate business scenarios

Stock Market Outperformance Prediction System Oct 2025 – Feb 2026
(Python, Pandas, scikit-learn) GitHub
- Built a data-driven prediction pipeline to identify stocks likely to outperform the S&P500 using historical market data
- Developed preprocessing pipelines for large-scale financial time-series data, aligning inconsistent sources and handling missing values
- Engineered features from raw financial reports using regex-based extraction and statistical indicators
- Implemented a Random Forest model with backtesting framework to evaluate performance under historical market conditions

Personal Site

Developer Portfolio & Project Dashboard Sep 2025 – Current
(HTML, TailwindCSS, JavaScript, React, Node.js, Git, Vercel) GitHub
- Built platform aggregating live backend data to showcase systems, analytics pipelines, and metrics
- Designed modular frontend components interfacing with multiple APIs and services

Technical Skills

Languages: Python, Java, C, C++, SQL, JavaScript, Go
Backend & Systems: FastAPI, Node.js, Express.js, REST APIs, API Design, Algorithms, data structures
Frontend: React, Next.js, TailwindCSS, HTML, CSS
Databases: PostgreSQL, MySQL, Database Design, Query Optimization
Tools & Infrastructure: Git, GitHub, Linux/Unix, Command Line, Docker (basic)
Data & Machine Learning: Pandas, NumPy, scikit-learn, PyTorch, Data Processing, ETL Pipelines

Education & Awards

University of Waterloo, Waterloo, Ontario Sep 2025 – Present
Honours Computer Science (Co-op) (Relevant Coursework: Designing Functional Programs, Elementary Algorithm Design and Data Abstraction, Tools and Techniques for Software Development, Object-Oriented Software Development, Linear Algebra, Cognitive Science, Statistics)

Canadian Open Mathematics Challenge (COMC) 2024: Honorable Mention; Best in Grade 12 (Central Toronto), Score: 66
Recognizes strong mathematical reasoning, proof skills, algorithmic thinking, and advanced problem solving
Euclid Mathematics Contest 2025: School Champion; Honor Roll, Score: 91/100 (Top 100 nationally)